Course Work Assessment - Alternative Assessment

Semester	202510	**Division**	CIS
Assessment title in Syllabus	AS Project	**Program**	Data Science

Course Code	CDS 3523		
Course Title	Statistical Inference	CLOs: 1 and 5	
Course Instructor	Asif Malik	**CRN**	14355
Assessment Weight	Group Project Report 50% + Oral Presentation (20%) + Written Defense (30%)	**Submission Date**	**Week 15/16**
For Group Work submissions, an additional individual assessment will be conducted. Grades for the students in one group will vary based on individual performance in the additional assessment.			

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقًا لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الإلتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبيّن وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض الى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

Student (s) Signature: ________________________

____

Student Name(s):	*Maryam Ali*	*Hessa Khalfan*	*Nourah Abdulla*
Student HCT ID(s):	H00535836	H00535166	H00535834

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Section No.				**Total**	**%**
Marks Allocated					
Marks Obtained					

Contents

1 – INTRODUCTION AND RESEARCH QUESTION FORMULATION

Introduction to the Education Sector

The **education** sector has undergone a significant transformation in recent years, largely driven by the adoption of digital technologies. The COVID-19 pandemic accelerated this shift, forcing institutions worldwide to adopt online and hybrid learning models to ensure educational continuity (UNESCO, 2020). Platforms like Microsoft Teams and Blackboard became essential for delivering lectures, managing coursework, and facilitating remote learning, with their usage becoming a well-documented aspect of the pandemic-era university experience (Al-Maroof et al., 2021). As institutions have returned to on-campus activities, the "hybrid" model has emerged as a popular approach, combining the benefits of face-to-face interaction with the flexibility of online learning.

This flexibility is a key advantage, allowing students to balance their studies with other commitments. Furthermore, research suggests that online learning can be highly effective. Some reports indicate that e-learning has the potential to increase information retention rates significantly compared to traditional face-to-face instruction, with some contexts showing retention improvements of up to 25-60% (Li, 2023). However, student engagement with digital resources is not uniform across the student body. A study on UK university students found that students in later years of study, for instance, tend to be less reliant on university-provided digital learning resources compared to first-year students, possibly due to increased confidence in their own learning strategies (Kantar, 2022).

Identifying Research Variables: This research aims to investigate the relationship between two key variables in the current academic environment:

Preferred Learning Mode: A categorical (nominal) variable describing the method students prefer for their courses. The options include In-person, Hybrid, or Online.

Year of Study: A categorical (ordinal) variable representing the student's academic level, from 1st Year to 4th Year or more.

Research Question: Based on the above, the research question is: **"Is there a statistically significant association between the preferred learning mode (in-person, hybrid, online) and the year of study among computer science students at the Higher Colleges of Technology?"**

This question is important because it helps to understand if students' learning preferences change as they progress through their academic journey. The findings from this research could provide valuable insights for educators and institutions to design academic programs that better meet the evolving needs of students at different stages.

2- SURVEY DESIGN

Survey Instrument

To collect the necessary data for this study, a survey was designed consisting of 10 relevant questions. The survey was structured to include a mix of both categorical variables and continuous variables.

Variable Definition and Measurement Scales

Each question in the survey was classified according to its measurement scale to ensure the correct application of statistical analysis. The table below defines each variable and its corresponding scale:

Table 1: Definition of Survey Variables and Measurement Scales

Q#	Question	Variable Type	Measurement Scale
1	Which age group do you belong to?	Categorical	Ordinal
2	Which division of computer science do you belong?	Categorical	Nominal
3	What is your current year of study?	Categorical	Ordinal
4	Which mode do you prefer for most of your courses?	Categorical	Nominal
5	Roughly, how much time per week do you spend on academic activities?	Continuous	Ratio
6	Have you used YouTube for academic purposes?	Categorical	Nominal
7	On average, how many minutes per day do you spend on Blackboard?	Categorical	Ordinal
8	Do you typically attend optional review sessions?	Categorical	Nominal
9	How would you rate the effectiveness of your primary learning mode (1-5)?	Continuous	Interval
10	On a scale of 0 to 10, how satisfied are you with your overall learning experience?	Continuous	Interval

Fulfillment of Inferential Testing Requirements

The survey was specifically designed to get the minimum number of variables needed for the chosen statistical test. **The Chi-square test** of independence requires two categorical variables. This requirement is met by providing "Year of Study" (Q3) and "Preferred Learning Mode" (Q4) in the survey.

Survey Form Creation

We created our survey using **Google Forms**, an online survey tool. This tool was chosen for its simplicity, the fact that participants can easily navigate through it, and its efficiency in collecting data, which automatically organize responses into a spreadsheet format (Microsoft Excel).

3- SAMPLING TECHNIQUES

Target Population

The target population for this research project is defined as all undergraduate students enrolled in the Computer Science division, which includes both the Information Technology (IT) and Information Systems (IS) programs, at the Higher Colleges of Technology.

Sampling Method and Sample Size

A **non-probability convenience sampling method** was employed for this study. To effectively reach the target population, the online survey link was distributed through digital communication channels and online student communities dedicated to HCT IT and IS students. This strategy aimed to maximize the recruitment of participants relevant to the research scope.

A total of 53 valid responses were collected through this method. This sample size successfully meets the minimum requirement of 50 participants as mentioned in the project guidelines.

Justification and Limitations

This sampling strategy was selected for its practical efficiency and its ability to directly access the intended student body. Utilizing established digital student communities was a pragmatic approach to gather data within the project's timeframe.

However, the limitations of this non-random sampling method must be acknowledged. Since participation was voluntary, the sample may not be perfectly representative of the entire IT and IS student population. For example, it might over-represent students who are more digitally active. Therefore, while the findings of this study provide valuable preliminary insights, they should be interpreted with consideration for the nature of the sampling technique.

4-DATA PREPARATION [CLO1]

Data Preparation

The initial dataset, comprising 53 responses from the "Student Learning Experience Survey," was imported for analysis. The primary objective of this phase was to ensure the data's integrity and prepare it for statistical testing.

The data preparation process included the following steps:

- **Data Cleaning:** The dataset was first inspected for missing values and duplicate entries. No missing data was found in the key analytical columns, and no duplicate responses were identified.
- **Data Standardization:** To ensure consistency, the categorical data was standardized. The lengthy descriptions in the 'Mode' column were simplified to three distinct categories: 'In-person', 'Hybrid', and 'Online'.
- **Category Consolidation:** For the 'Year' variable, the two responses from students in "More than 4" years of study were merged with the "4th Year" category. This step was taken to strengthen the validity of the Chi-square test by ensuring adequate cell counts.

Following these procedures, a clean and structured dataset of 53 responses was finalized for analysis.

Descriptive Statistics

Descriptive statistics were calculated to summarize the fundamental characteristics of the sample population. This analysis provides an overview of the distribution of students by academic year and their stated learning preferences.

The key findings from the descriptive analysis are as follows:

- The sample is predominantly composed of **3rd Year students**, who account for 67.9% of all participants (n=36).

- The **Hybrid** learning mode was the most frequently selected option, chosen by 49.1% of students (n=26), followed closely by the **In-person** mode at 43.4% (n=23). The fully **Online** mode was the least preferred, representing only 7.5% of the sample (n=4).
- The mean score for overall learning satisfaction was **6.74** on a 10-point scale, with a median of **7.0**, indicating a generally positive level of satisfaction among the respondents.

The detailed frequency distributions are presented in the tables below.

Table 2: Frequency Distribution of Respondents by Year of Study

Year of Study	Frequency (n)	Percentage (%)
3rd Year	36	67.9%
2nd Year	8	15.1%
4th Year	6	11.3%
1st Year	3	5.7%
Total	**53**	**100.0%**

Table 3: Frequency Distribution of Respondents by Preferred Learning Mode

Preferred Learning Mode	Frequency (n)	Percentage (%)
Hybrid	26	49.1%
In-person	23	43.4%
Online	4	7.5%
Total	**53**	**100.0%**

5- NORMALITY TESTING [CLO5]

Introduction to Normality Testing

Normality testing is a critical step in statistical analysis to determine whether a dataset follows a normal distribution (also known as a bell curve). This assumption is a prerequisite for many parametric statistical tests, such as the t-test and ANOVA. In this section, we assess the normality of the continuous variable 'Satisfaction_Rating' using a combination of visual methods and a formal statistical test, as required by the project guidelines.

Methods Used

To conduct a comprehensive assessment, the following three methods were employed:

1. **Histogram:** A graphical representation of the data's distribution. A bell-shaped curve suggests normality.
2. **Quantile-Quantile (Q-Q) Plot:** This plot compares the quantiles of the sample data against the quantiles of a theoretical normal distribution. If the data is normally distributed, the points will fall approximately along a straight diagonal line.
3. **Shapiro-Wilk Test:** A formal statistical test that provides a p-value to quantify the evidence against normality. A p-value less than 0.05 is typically interpreted as significant evidence that the data is not normally distributed.

Results and Interpretation

The normality tests were performed on the 'Satisfaction_Rating' variable (N=53).

Visual Inspection:

- **Histogram (Figure 5.1):** The histogram of satisfaction scores does not exhibit a clear bell shape. Instead, it shows multiple peaks, particularly around the scores of 6, 8, and 10, and appears to be skewed to the left.
- **Q-Q Plot (Figure 5.2):** The points on the Q-Q plot deviate from the straight reference line, especially at the lower and upper ends. This deviation indicates that the tails of the distribution are not consistent with those of a normal distribution.

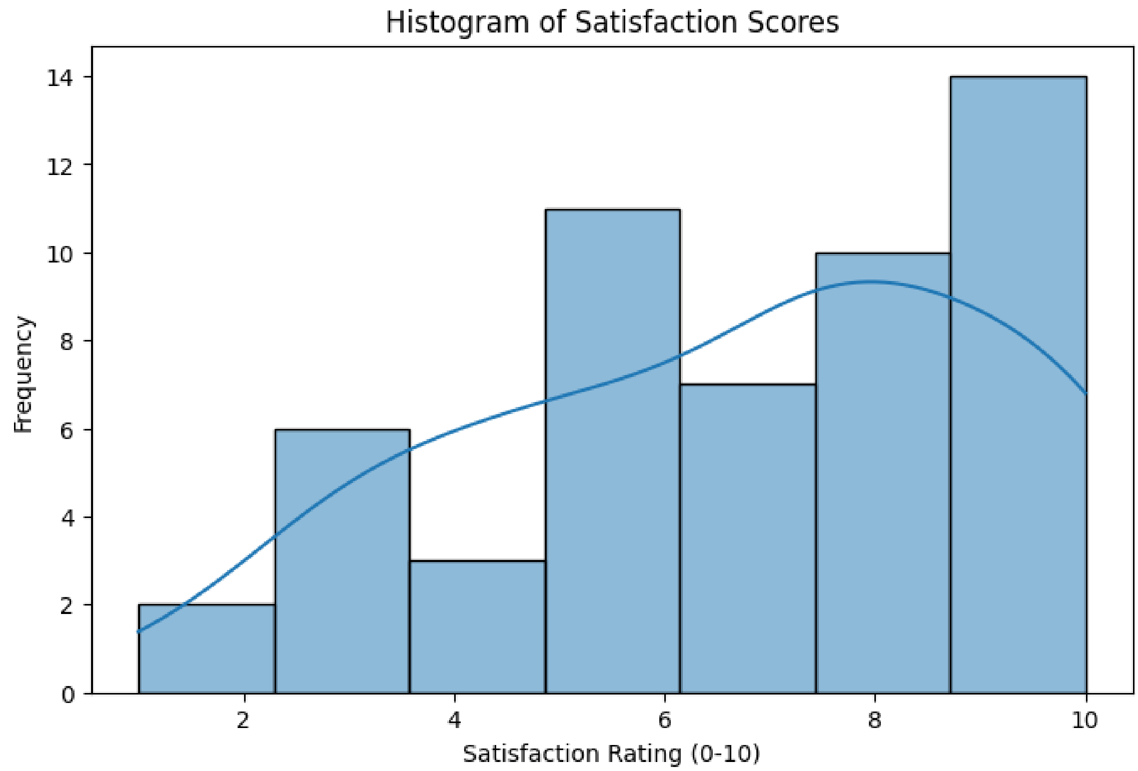

Figure 1: Histogram of Satisfaction Scores

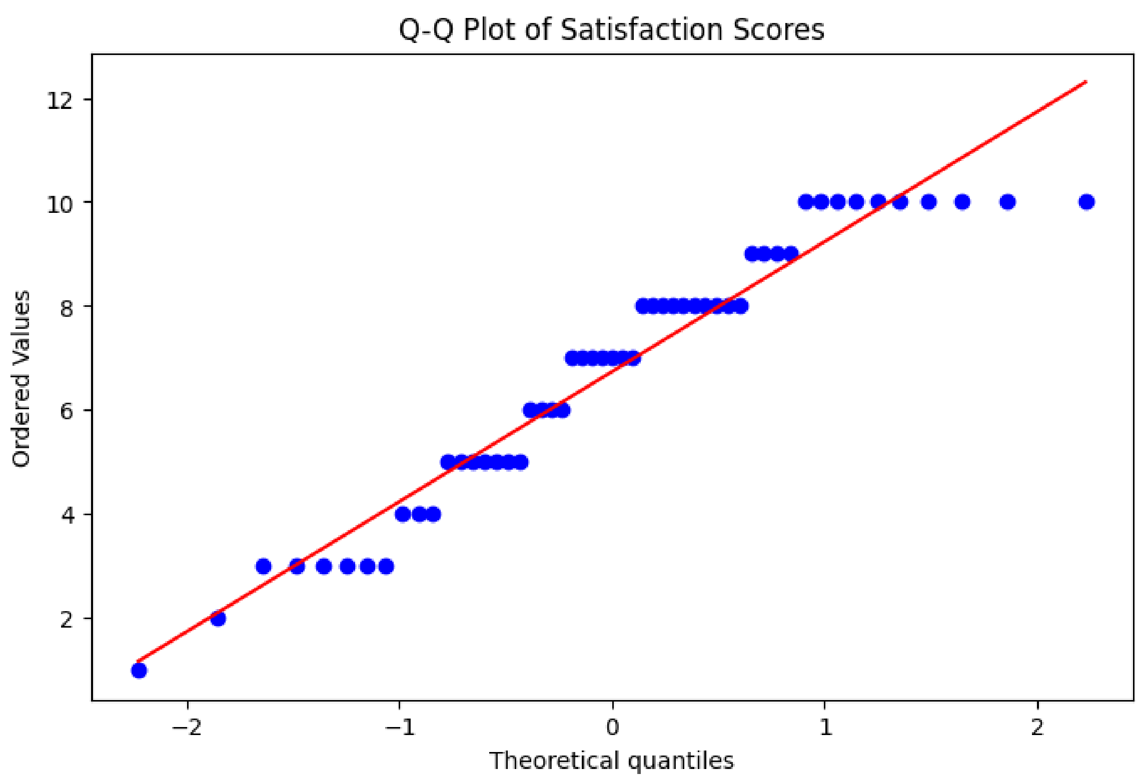

Figure 2: Q-Q Plot of Satisfaction Score

Statistical Test: The Shapiro-Wilk test was conducted, yielding the following result:

- **Shapiro-Wilk Statistic:** 0.925
- **P-value:** 0.004

Conclusion on Normality

The p-value obtained from the Shapiro-Wilk test is **0.004**, which is less than the standard alpha level of 0.05. This result, combined with the visual evidence from the histogram and the Q-Q plot, provides strong evidence to **reject the null hypothesis of normality**.

Therefore, we conclude that the 'Satisfaction_Rating' data in this sample **does not meet the assumption of normality**. This finding is important as it would require the use of non-parametric tests if this variable were to be used in further hypothesis testing that assumes normality.

6-STATISTICAL DATA ANALYSIS [CLO5]

Choice of Hypothesis Test The research question aims to determine if a statistically significant association exists between two categorical variables: 'Year of Study' and 'Preferred Learning Mode'. The appropriate statistical test for this objective is the **Chi-Square (χ^2) Test of Independence**. This test is designed to evaluate whether the observed frequencies in a contingency table differ significantly from the frequencies that would be expected if the two variables were independent.

6.2. Hypotheses, Significance Level, and Assumptions To formally test the research question, the following statistical framework was established:

- **Null Hypothesis (H_0):** There is no statistically significant association between a student's year of study and their preferred learning mode. The two variables are independent.
- **Alternative Hypothesis (H_1):** There is a statistically significant association between a student's year of study and their preferred learning mode. The two variables are dependent.
- **Significance Level (α):** The significance level, or alpha (α), was set at **0.05**. This is the standard threshold in social sciences, indicating a 5% risk of concluding that an association exists when one does not.
- **Assumptions:** The Chi-square test assumes that each observation is independent and that the expected frequency for each cell in the contingency table is 5 or

greater. It was noted that some cells in this analysis had an expected frequency below 5, which is a limitation of the current sample size.

Hypothesis Testing and Results the Chi-square test was performed using Python's scipy.stats library. A contingency table was first generated to cross-tabulate the observed frequencies of the two variables.

Table 4: Contingency Table of Observed Frequencies for Year of Study vs. Preferred Mode

Year of Study	Hybrid	In-person	Online	**Total**
1st Year	1	1	1	**3**
2nd Year	3	4	1	**8**
3rd Year	20	15	1	**36**
4th Year	2	3	1	**6**
Total	**26**	**23**	**4**	**53**

The test yielded the following results:

- **Chi-Square (χ^2) Statistic:** 4.54
- **Degrees of Freedom (df):** 6
- **P-value:** 0.604

6.4. Interpretation of Results The p-value obtained from the Chi-square test is **0.604**. To make a conclusion, this value is compared to the pre-defined significance level (α = 0.05).

Since the p-value (0.604) is substantially greater than alpha (0.05), we **fail to reject the null hypothesis (H_0)**.

This outcome indicates that there is **no statistically significant evidence** within this sample to suggest an association between a student's year of study and their preferred learning mode. The variations in preferences observed across the different year levels (as shown in Figure 6.1) are likely attributable to random chance rather than a true, underlying relationship. In other words, a student's preference for in-person, hybrid, or online learning does not appear to depend on their academic year.

Figure 6.1: Bar Chart of Preferred Learning Mode by Year of Study *(This chart visually supports the conclusion by showing no clear or consistent pattern of preference as*

students advance in their studies. For instance, the preference for 'Hybrid' does not uniformly increase or decrease across the year levels.)

CONCLUSION AND RECOMMENDATIONS

Summary of Key Findings

This research project was undertaken to investigate the learning preferences of computer science students at the Higher Colleges of Technology. The primary research question focused on determining whether a statistically significant association exists between a student's year of study and their preferred learning mode (in-person, hybrid, or online).

The key findings from the statistical analysis are summarized as follows:

1. **Descriptive Findings:** The analysis of the 53 responses revealed that the sample was predominantly composed of **3rd Year students (67.9%)**. The **Hybrid learning mode was the most popular choice (49.1%)**, followed closely by the In-person mode (43.4%). Full Online mode was the least preferred option. The overall student satisfaction with their learning experience was generally positive, with a mean score of 6.74 out of 10.

2. **Normality Testing:** A normality test was conducted on the 'Satisfaction_Rating' variable. The Shapiro-Wilk test yielded a p-value of **0.004**, which is less than the alpha level of 0.05. This result, supported by visual inspection of a histogram and a Q-Q plot, led to the conclusion that the satisfaction data **is not normally distributed**.

3. **Hypothesis Testing:** The core of the analysis was the Chi-Square Test of Independence, which was used to test the relationship between 'Year of Study' and 'Preferred Learning Mode'. The test resulted in a **p-value of 0.604**. Since this value is significantly greater than the 0.05 significance level, we **failed to reject the null hypothesis**.

Conclusion

The primary conclusion of this study is that there is **no statistically significant association** between a student's year of study and their preferred learning mode within this sample. The observed differences in preferences across the various year levels are likely due to random chance rather than a systematic pattern. Therefore, based on our data, we cannot conclude that students' learning preferences change as they progress through their academic program.

Actionable Insights and Recommendations

While the main hypothesis was not supported, the descriptive data provides several actionable insights for the institution:

- **Dominance of Hybrid and In-person Learning:** The strong preference for both Hybrid (49.1%) and In-person (43.4%) modes suggests that students highly value flexibility combined with face-to-face interaction. The very low preference for the fully Online mode (7.5%) indicates that a complete shift to remote learning may not be well-received by the student body.
  - **Recommendation 1:** The college should continue to invest in and refine its **Hybrid learning model**. This could involve training instructors on best practices for blended learning and ensuring that the technological infrastructure is robust and reliable.
- **Student Satisfaction:** A mean satisfaction score of 6.74, while positive, indicates that there is room for improvement.
  - **Recommendation 2:** It is recommended that the administration conduct further qualitative research (e.g., focus groups, open-ended feedback) to understand the specific factors that are impacting student satisfaction. This could help identify areas for enhancement in curriculum, teaching methods, or student support services.

Limitations and Future Research

It is important to acknowledge the limitations of this study, which can guide future research on this topic:

- **Sample Composition:** The most significant limitation was the unbalanced nature of the sample. The dataset was heavily skewed towards 3rd Year students, while 1st and 2nd Year students were underrepresented. This imbalance may have masked a true relationship between the variables and limits the generalizability of the findings.

  Future Research 1: Future studies should employ a **stratified sampling technique** to ensure a balanced and representative sample from each academic year. This would provide a more accurate basis for comparison.

- **Sample Size:** A sample size of 53 is relatively small. A larger sample would increase the statistical power of the analysis and provide more reliable results.

  Future Research 2: It is recommended that future research aim for a larger sample size to increase confidence in statistical conclusions.

Python Codes

The Python codes used for data cleaning, descriptive statistics, normality testing, and the Chi-Square test are provided in **Appendix B** of this report.

APPENDIX A: SURVEY QUESTIONS

1-Which age group do you belong to?

- 17 or younger
- 18 - 20 years old
- 21 - 23 years old
- 24 - 26 years old
- 27 or older

2- Which division of computer science do you belong?

- Information Technology (IT)
- Information Systems (IS)

3- What is your current year of study?

- 1st Year
- 2nd Year
- 3rd Year
- 4th Year
- More than 4

4- Which mode do you prefer for most of your courses?

- In-person (traditional classroom setting)
- Hybrid (mix of in-person and online)
- Online (fully remote)

5- Roughly, how much time per week do you spend on academic activities (lectures, studying, assignments)?

Open ended question

6-Have you used YouTube for academic purposes (e.g., watching tutorials, explanations) in the past month?

- Yes
- No

7- On average, how many minutes per day do you spend on Blackboard?

- Less than 15 minutes
- 15 - 30 minutes
- 31 - 60 minutes

- more than 60 minutes

8- Do you typically attend optional review sessions or tutorials?

- Yes
- No

9- How would you rate the effectiveness of your primary learning mode (chosen in Q4) on a scale of 1 to 5? (1 = Very Ineffective, 5 = Very Effective)

1, 2, 3, 4, 5

10- On a scale of 0 to 10, how satisfied are you with your overall learning experience this academic year? (0 = Not at all satisfied, 10 = Extremely satisfied)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

APPENDIX B: PYTHON CODE

```
# Step 1: Import the necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# --------------------------------------------------------------------
------------------

# Step 2: Load the data from the Excel file
file_name = 'Student Learning Experience Survey (Responses).xlsx'
df = pd.read_excel(file_name)
```

```
# Step 3: Clean the data
# Rename ALL column names to shorter, easier-to-use names
df.rename(columns={
    '1- Which age group do you belong to?': 'Age_Group',
    '2- To which division of computer science do you belong ?':
'Division',
    '3- What is your current year of study?': 'Year',
    '4- Which mode do you prefer for most of your courses?': 'Mode',
    '5- Roughly, how much time per week do you spend on academic
activities (lectures, studying, assignments)?': 'Time_Spent_Weekly',
    '6-Have you used YouTube for academic purposes (e.g., watching
tutorials, explanations) in the past month?': 'Used_YouTube',
    '7- On average, how many minutes per day do you spend on
Blackboard?': 'Blackboard_Time_Daily',
    '8- Do you typically attend optional review sessions or
tutorials?': 'Attend_Optional_Sessions',
    '9- How would you rate the effectiveness of your primary
learning mode (chosen in Q4) on a scale of 1 to 5? (1 = Very
Ineffective, 5 = Very Effective)': 'Effectiveness_Rating',
    '10- On a scale of 0 to 10, how satisfied are you with your
overall learning experience this academic year? (0 = Not at all
satisfied, 10 = Extremely satisfied)': 'Satisfaction_Rating'
}, inplace=True)

# Standardize the text in the 'Mode' column
df['Mode'] = df['Mode'].apply(lambda x: 'In-person' if 'In-person'
in x else ('Hybrid' if 'Hybrid' in x else 'Online'))
```

```python
# Consolidate the 'More than 4' category into '4th Year'
df['Year'] = df['Year'].replace({'More than 4': '4th Year'})

# Step 4: Calculate and display descriptive statistics
print("--- 2. Descriptive Statistics ---")

# Table for 'Year of Study' distribution
year_counts = df['Year'].value_counts()
print("Distribution of Students by Year of Study:")
print(year_counts)
print("-" * 40)

# Table for 'Preferred Learning Mode' distribution
mode_counts = df['Mode'].value_counts()
print("\nDistribution of Students by Preferred Learning Mode:")
print(mode_counts)
print("-" * 40)

# Calculate Mean and Median for 'Satisfaction_Rating'
mean_satisfaction = df['Satisfaction_Rating'].mean()
median_satisfaction = df['Satisfaction_Rating'].median()
print(f"\nMean Satisfaction Rating: {mean_satisfaction:.2f} out of
10")
print(f"Median Satisfaction Rating: {median_satisfaction} out of
10")
```

```python
# Step 4: Calculate and display descriptive statistics
print("--- 2. Descriptive Statistics ---")

# Table for 'Year of Study' distribution
year_counts = df['Year'].value_counts()
print("Distribution of Students by Year of Study:")
print(year_counts)
print("-" * 40)

# Table for 'Preferred Learning Mode' distribution
mode_counts = df['Mode'].value_counts()
print("\nDistribution of Students by Preferred Learning Mode:")
print(mode_counts)
print("-" * 40)

# Calculate Mean and Median for 'Satisfaction_Rating'
mean_satisfaction = df['Satisfaction_Rating'].mean()
median_satisfaction = df['Satisfaction_Rating'].median()
print(f"\nMean Satisfaction Rating: {mean_satisfaction:.2f} out of 10")
print(f"Median Satisfaction Rating: {median_satisfaction} out of 10")
```

```
--- 2. Descriptive Statistics ---
Distribution of Students by Year of Study:
Year
3rd Year    36
2nd Year     8
4th Year     6
1st Year     3
Name: count, dtype: int64
----------------------------------------

Distribution of Students by Preferred Learning Mode:
Mode
Hybrid       26
In-person    23
Online        4
Name: count, dtype: int64
----------------------------------------

Mean Satisfaction Rating: 6.74 out of 10
Median Satisfaction Rating: 7.0 out of 10
```

```python
# --- A. Normality Test for 'Satisfaction_Rating' ---

satisfaction_data = df['Satisfaction_Rating']

# Plot 1: Histogram
plt.figure(figsize=(8, 5))
```

```
sns.histplot(satisfaction_data, kde=True)
plt.title('Histogram of Satisfaction Scores')
plt.xlabel('Satisfaction Rating (0-10)')
plt.ylabel('Frequency')
plt.show()
```

```
# --- A. Normality Test for 'Satisfaction_Rating' ---
satisfaction_data = df['Satisfaction_Rating']

# Plot 1: Histogram
plt.figure(figsize=(8, 5))
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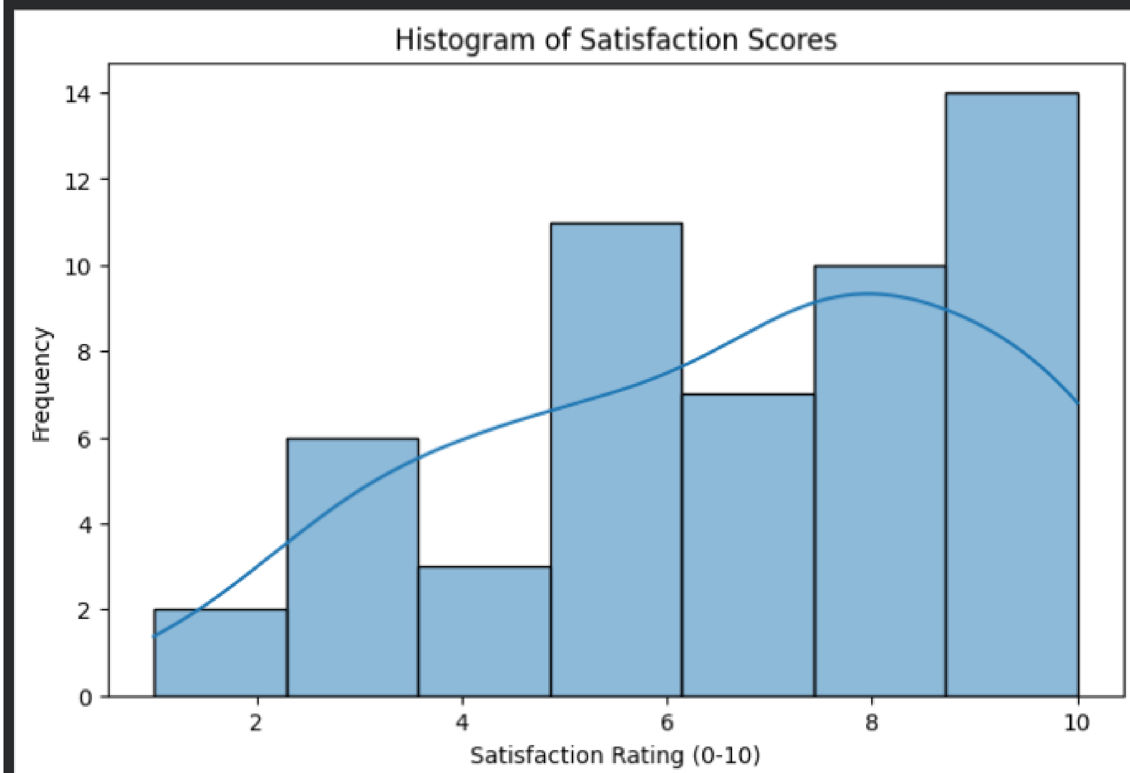


```
# Plot 2: Q-Q Plot

plt.figure(figsize=(8, 5))
stats.probplot(satisfaction_data, dist="norm", plot=plt)
plt.title('Q-Q Plot of Satisfaction Scores')
plt.show()
```

```python
# Plot 2: Q-Q Plot
plt.figure(figsize=(8, 5))
stats.probplot(satisfaction_data, dist="norm", plot=plt)
plt.title('Q-Q Plot of Satisfaction Scores')
plt.show()
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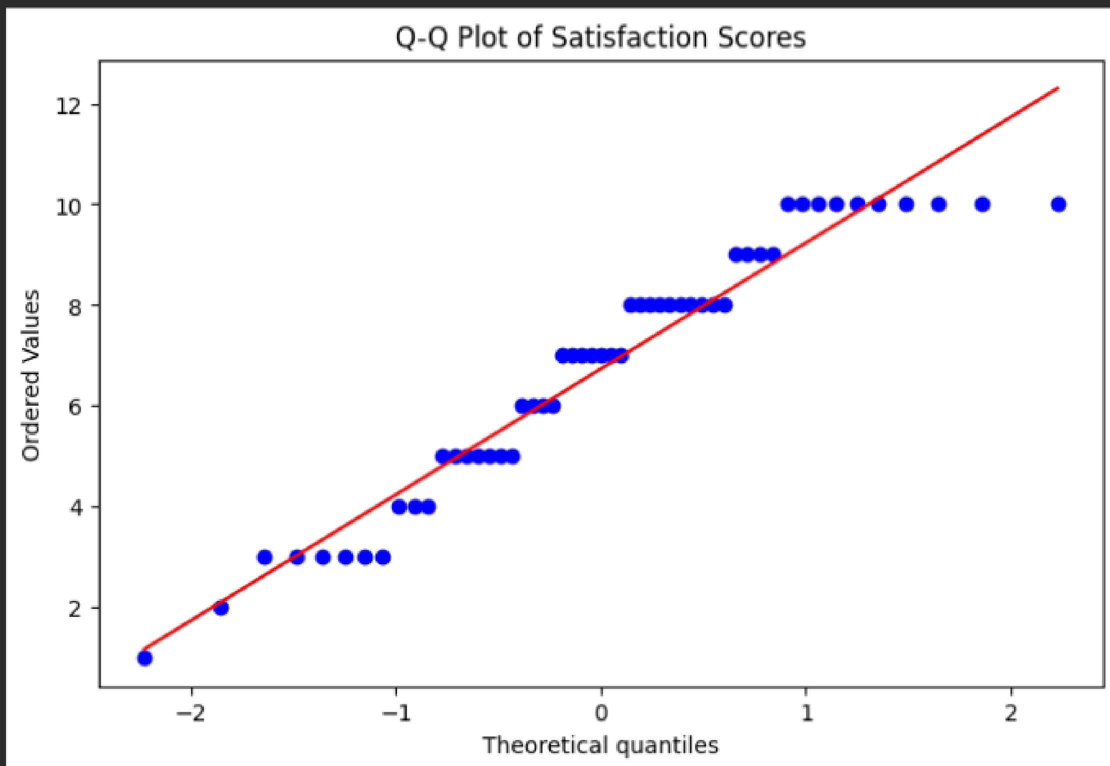


```python
# Statistical Test: Shapiro-Wilk

stat, p_value = stats.shapiro(satisfaction_data)
print(f"Shapiro-Wilk Test Results:")
print(f"P-value: {p_value:.3f}")
if p_value > 0.05:
    print("Conclusion: The data appears to be normally distributed
(fail to reject H0).")
else:
    print("Conclusion: The data does not appear to be normally
distributed (reject H0).")
```

```python
# Statistical Test: Shapiro-Wilk
stat, p_value = stats.shapiro(satisfaction_data)
print(f"Shapiro-Wilk Test Results:")
print(f"P-value: {p_value:.3f}")
if p_value > 0.05:
    print("Conclusion: The data appears to be normally distributed (fail to reject H0).")
else:
    print("Conclusion: The data does not appear to be normally distributed (reject H0).")
```

```
Shapiro-Wilk Test Results:
P-value: 0.004
Conclusion: The data does not appear to be normally distributed (reject H0).
```

```python
# --- B. Chi-Square Test of Independence ---

print("--- 4. Chi-Square Test (Year of Study vs. Preferred Mode) ---
")


# Create a contingency table (cross-tabulation)

contingency_table = pd.crosstab(df['Year'], df['Mode'])

print("Contingency Table (Observed Frequencies):")

print(contingency_table)

print("-" * 40)


# Perform the Chi-Square test

chi2, p, dof, expected = stats.chi2_contingency(contingency_table)

print(f"\nChi-Square Test Results:")

print(f"P-value: {p:.3f}")


if p < 0.05:

    print("Conclusion: There is a statistically significant
association between Year of Study and Preferred Mode (reject H0).")

else:

    print("Conclusion: There is no statistically significant
association between Year of Study and Preferred Mode (fail to reject
```

```
H0).")
```

```python
# --- B. Chi-Square Test of Independence ---
print("--- 4. Chi-Square Test (Year of Study vs. Preferred Mode) ---")

# Create a contingency table (cross-tabulation)
contingency_table = pd.crosstab(df['Year'], df['Mode'])
print("Contingency Table (Observed Frequencies):")
print(contingency_table)
print("-" * 40)

# Perform the Chi-Square test
chi2, p, dof, expected = stats.chi2_contingency(contingency_table)
print(f"\nChi-Square Test Results:")
print(f"P-value: {p:.3f}")

if p < 0.05:
    print("Conclusion: There is a statistically significant association between Year of Study and Preferred Mode (reject H0).")
else:
    print("Conclusion: There is no statistically significant association between Year of Study and Preferred Mode (fail to reject H0).")
```

```
--- 4. Chi-Square Test (Year of Study vs. Preferred Mode) ---
Contingency Table (Observed Frequencies):
Mode      Hybrid  In-person  Online
Year
1st Year       1          1       1
2nd Year       3          4       1
3rd Year      20         15       1
4th Year       2          3       1
----------------------------------------

Chi-Square Test Results:
P-value: 0.438
Conclusion: There is no statistically significant association between Year of Study and Preferred Mode (fail to reject H0).
```

```python
# Perform the Chi-Square test

chi2, p, dof, expected = stats.chi2_contingency(contingency_table)

print(f"\nChi-Square Test Results:")

print(f"P-value: {p:.3f}")


if p < 0.05:

    print("Conclusion: There is a statistically significant association between Year of Study and Preferred Mode (reject H0).")

else:

    print("Conclusion: There is no statistically significant association between Year of Study and Preferred Mode (fail to reject H0).")
```

```
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chi2, p, dof, expected = stats.chi2_contingency(contingency_table)
print(f"\nChi-Square Test Results:")
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```

```
Chi-Square Test Results:
P-value: 0.438
Conclusion: There is no statistically significant association between Year of Study and Preferred Mode (fail to reject H0).
```

```
# Plot 3: Bar chart to visualize the relationship

contingency_table.plot(kind='bar', figsize=(10, 6), rot=0)

plt.title('Preferred Learning Mode by Year of Study')

plt.xlabel('Year of Study')

plt.ylabel('Number of Students')

plt.legend(title='Preferred Mode')

plt.show()
```


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contingency_table.plot(kind='bar', figsize=(10, 6), rot=0)
plt.title('Preferred Learning Mode by Year of Study')
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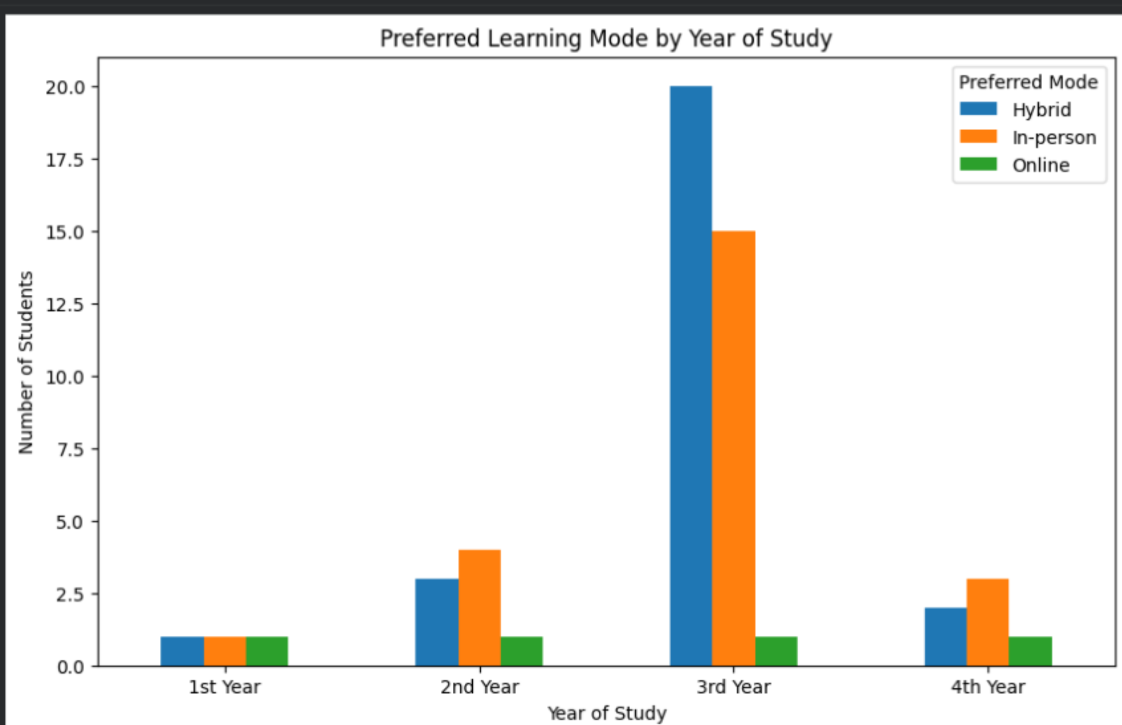
```

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